Urban Heat Prediction And Mapping

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***Abstract*—The increasing rate of urbanization and changes in land use pose significant environmental challenges, including the Urban Heat Island (UHI) effect, which causes an increase in land surface temperature (LST) in affected urban areas due to built-up area expansion, reduced vegetation cover, and increased human activities.**

As such, accurate prediction and spatial mapping of urban heat are critical to implement effective urban design and mitigation.

In this paper, we provide an integrated approach to predicting urban heat and creating urban heat maps that employ remote sensing data, geospatial analysis and machine learning methods. Key parameters such as land use/land cover, vegetation indices, built-up indices, and climate factors were examined for their effects on temperature variation; machine learning models were used to predict land surface temperatures, and Geographic Information System (GIS) tools were used to create spatial heat maps to identify areas of urban heat concentration.This study shows that an enhanced prediction accuracy, the clear visualization of heat distributions in urban areas with respect to urban morphologies/environmental features affects the temperature patterns, and supports data driven decision making for sustainable urban development and efficient heat mitigation planning is fully utilized through this research.

***Index Terms*—Urban Heat Island , Land Surface Temperature , Remote Sensing, Geographic Information System , Machine Learning, Urban Heat Prediction, Heat Mapping, Land Use/Land Cover**

I. Introduction

The high rate of urbanization has greatly changed the natural landscapes into built-up environments, which have resulted in a number of environmental challenges. Among the most serious problems, the rise in temperature in cities in comparison to the surrounding rural areas can be mentioned, which is known as the Urban Heat Island (UHI) effect. The factors that primarily lead to this phenomenon include dense infrastructure, less vegetation cover, more impervious surfaces and anthropogenic heat emissions.

Urban heat has serious impacts on human health, energy consumption and environmental sustainability. Studies have shown that land use and land cover (LULC) changes play a major role in increasing land surface temperature (LST) in cities. As an example, seasonal changes and vegetation cover are strong factors that affect the thermal conditions in urban areas, particularly developing ones like India [1]. On the same note, extensive urban growth has been cited as a primary cause of the increased urban temperatures and the increased effects of UHI [2].

As such, traditional surface temperature measurement techniques using weather stations are limited in their ability to capture the spatial heterogeneity present across urban spaces. Hence, remote sensing and geospatial methods are extensively used to assess and map the land surface temperature. Remote sensing from satellites offers an effective and precise means of investigating temperature distribution to determine urban heat island hot spots [3].

Machine learning methods have gained momentum in the last decades. Data-driven approaches like regression and artificial neural networks (ANN) have been found to be effective in estimating land surface temperature with high accuracy [4]. Other AI-based advanced machine learning approaches such as random forest have also been applied to urban heat severity analysis and the identification of influencing factors [5], [6].

Brilliantly, there have been some research works to understand the factors that enforce urban heat. Among others urban morphology, spatial structure and built-up density have been named important factors driving intra-urban temperature heterogeneity [7], [8]. Further, longitudinal research indicates that urbanization results in temperature trends with significant increases occurring over spans of many years, necessitating predictive modeling and sustainable management to ameliorate this effect [9].When it comes to understanding urban heat, researchers have made some real progress, but there's still a lot to learn. Most studies so far have focused on either predicting urban heat or analyzing it, but not really combining the two. What we really need is a way to bring together machine learning, which can help us predict urban heat, and spatial mapping, which can show us where the heat is. By doing this, we can get a much better understanding of how

urban heat is distributed. That's why this study is trying to create a comprehensive framework that uses geospatial data and advanced analytical techniques to predict and map urban heat. This approach will help us get a clearer picture of urban heat and how it affects our cities.

II. Related work

Remote sensing, geospatial, and machine learning methods have been employed to pervasively study urban heat and land surface temperature (LST). The primary emphasis of existing studies is on temperature prediction, analysis of the influencing factors, and spatial evaluation of the effects of urban heat island (UHI).

A number of scholars have examined machine learning methods to predict urban heat. Gupta and Aithal [4] constructed a data-driven model with multiple linear regression and artificial neural networks (ANN) to predict land surface temperature, and they found better accuracy of ANN models. In a similar fashion, Mansouri and Erfani [5]used machine learning algorithms to forecast the severity of urban heat island, which demonstrated the usefulness of data-driven methods in determining high-risk areas. Land use and land cover (LULC) alterations are considered to be one of the significant sources of urban heat. Talukdar et al. [1] studied the seasonal changes in LST and demonstrated that vegetation has a significant effect in mitigating urban temperature. Li et al. [2] also pointed out the fact that high urbanization and land conversion exacerbate the UHI and raise the surface temperature.

Spatial analysis of urban heat has been done widely through remote sensing methods and GIS-based methods. The application of satellite images in estimating LST and detecting temperature differences in urban centers was demonstrated by Cetin et al. [3]. Continuous urbanization, as it is observed by long-term projects like Liyaqat et al. [9] causes enduring rises in temperature and predictive analysis is required. Besides prediction, a number of studies have aimed to determine the key drivers of urban heat. He et al. [7]employed the random forest models to determine the effect of urban spatial structure on the temperature distribution, whereas Feng et al. [8] determined the effect of environmental and structural elements on the LST. Another study by Huang et al. [10] also demonstrated that urban morphology is a major factor that influences patterns in UHI.

The latest developments in artificial intelligence have contributed to the improvement of urban heat studies. Merabtine and Snaiki [11] and Ahmed et al. [12] conducted reviews of different machine learning techniques in UHI applications, noting their increasing value. Hoang et al. [6] used geospatial analysis and interpretable machine learning to understand better the correlation between environmental features and changes in temperature.

In spite of these developments, a majority of existing studies concentrate on either prediction or spatial analysis alone. Minimal studies combine both machine learning-based forecasting and GIS-based heat mapping into a single platform. Thus, the proposed study will help to create a combined method of urban heat forecast and spatial mapping to enhance knowledge and promote sustainable urban planning.

III. Methodology

A. Data Acquisition and Preprocessing

Using multi-source datasets from remote sensing platforms, climatic record and geospatial databases, this study geo-analyses. Urban heat analysis requires Land Surface Temperature (LST) data obtained from satellite images, like Landsat, which have thermal infrared channels. It further includes auxiliary datasets such as land use/land cover (LULC) or vegetation indices, built-up indices, and other environmental variables since they greatly affect the urban thermal characteristics [4] [1] [2].

- Filtering out noise and handling missing values
- Remote sensing satellite image geo-corrections
- Feature normalization and scaling
- Joining and resampling of lines for datasets to common resolution

  These activities guarantee that the heterogeneous datasets are homogenous and ready for next sequential processing. Derived indices from remote sensing are calculated during pre-processing to enhance the representation of features [3].

B. Feature Extraction and Model Development

The variables that affect urban heat trends would have been identified through feature extraction techniques. The most important features are as listed below:

- The Normalized Difference Vegetation Index (NDVI), which provides a measure of the vegetation's coverage and thus its cooling capacity.
- The Normalized Difference Built-up Index (NDBI), which represents the degree of urbanization.
- Land surface temperature (LST), which is the dependent variable for prediction.
- Land use and/or Land cover (LULC), which shows the type of land present and classes of urban areas.

  These four have been chosen based on data collected from other published works illustrating how vegetation lowers heat and impervious areas contribute to higher heat [1], [7], [8].

  To establish the relationship between these four features and temperature, multiple machine learning algorithms have used to develop the following relationships:
- Multiple Linear Regression (MLR) is used as the baseline model.
- Random Forest (RF) can capture nonlinearity and importance of features.
- Artificial Neural Networks (ANN) can model complex patterns and provide high prediction accuracy.

  Feature extraction is performed to derive meaningful parameters from the datasets, such as vegetation cover and built-up density, which significantly influence urban heat patterns.

Multiple models, including regression and Random Forest, are implemented and comparatively analyzed to evaluate their performance in predicting land surface temperature (LST). The comparison is based on standard evaluation metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination (R^2). The dataset is divided into training and testing sets to ensure proper model validation and avoid overfitting.

The trained models are then used to predict temperature values based on user-defined inputs such as location and year. The models were trained using the same datasets that have been developed from the historic datasets and were validated using appropriate means of splitting the datasets. Hyper-parameter optimization was conducted on the models to maximize their performance. Previous studies demonstrated the superiority of Random Forest (RF) and Artificial Neural Networks (ANN) over conventional approaches to urban heat prediction [4], [5] [6].

TABLE I
FEATURES USED FOR URBAN HEAT PREDICTION

S.No	Feature	Description
1	NDVI	Vegetation Index
2	NDBI	Built-up Index
3	LST	Land Surface Temperature
4	LULC	Land Use/Land Cover

C. *Prediction, Spatial Mapping, and Evaluation*

After training, the temperature models use the trained model to predict the land surface temperatures (LST) for the area being studied. The method for predicting land surface temperature includes combining a variety of environmental/spatial data into the model to produce temperature estimates for specific locations.

The predicted land surface temperatures are then entered into a Geographic Information System (GIS) to create mapped visualizations of the land surface temperature. The mapped visualizations show the area to be classified into three different thermal zones an urban environment based on predicted land surface temperature:

- Hot Zones (i.e., Hot Spots)
- Moderate Zones
- Cool Zones

Spatial visualizations show the urban heat distribution, allowing urban planners to utilize the maps to assist in making decisions for future mitigation strategies [3], [9]. Model performance is measured using Statistical Techniques:

- Mean Absolute Error (MAE) — A statistical metric used to evaluate the accuracy of models.
- Root Mean Square error (RMSE)
- Coefficient of Determination (R^2)

These are the three Standard Statistical Methods that will assess model reliability and accuracy of the predictions produced using this model. Once the models are predicted, they will be compared to each other to determine which model has produced the most accurate land surface temperature predictions.

IV. PROPOSED METHOD

In this work we combine statistical regression techniques with an ensemble-based Random Forest (RF) algorithm to provide a robust and integrated framework for urban heat prediction and classification. By leveraging both spatial and temporal information, the intent of the landmarking technique presented here is two-fold; in the first part, we target accurate modeling of land surface temperature (LST) values while, in second component, we seek to screen urban regions by defining various levels of heat intensity. Leveraging the ability of this model to analyze complex interaction between environmental, geo-spatial and time series data allow us a robust scalable solution for urban heat analysis. The architectural model of urban thermal environment prediction system can accept custom inputs (e.g., location, year), allowing areas and years not included in the training set to be predicted dynamically. Besides user inputs, it also includes several environmental parameters, such as land use/land cover (LULC), Normalized Difference Vegetation Index (NDVI) and Normalized Difference Built-up Index (NDBI) as well as historical LSTs. The parameters are examined as they exert a powerful effect on the variance of temperature in urban areas; while vegetation leads to effects of cooling, built-up areas increase heat accumulation. Regression techniques are used to create base relations between input parameters and temperature as first step. From here you learn the relationship between physical parameters and urban heat is linear However, due to the non linear complex nature of urban heat patterns a Random Forest algorithm is used as main predictive model. Random Forest (RF) is another ensemble learning approach, which consists of multiple decision trees that outputs outcomes are aggregated to enhance prediction accuracy while reducing variance and overfitting. Furthermore, it permits feature importance studies to figure out which factors play a significant role in creating urban heat.

- Acquisition of multi-source geospatial and environmental datasets
- Integration of user inputs (location and year) for dynamic prediction
- Data preprocessing, including noise removal, normalization, and spatial alignment
- Extraction of key features such as NDVI, NDBI, LULC, and LST
- Application of regression analysis for baseline modeling
- Implementation of Random Forest for advanced prediction and nonlinear modeling
- Training and validation of models using historical data
- Prediction of land surface temperature for selected inputs
- Classification of predicted temperature into heat intensity categories
- Visualization of results using color-coded spatial representation

The predicted temperature values are further classified into three distinct heat intensity levels to enhance interpretability. These include:

- High temperature (Hotspots) → represented in Red
- Medium temperature zones → represented in Orange
- Low temperature zones → represented in Green

This classification system also organizes the study of urban heat in a methodical manner and helps quickly pinpoint hot spot areas that require interventions. By using GIS-based modeling it is possible to generate accessible heat maps for a clear understanding of the spatial distribution of temperature in urbanization.

Moreover, the proposed approach allows decision-making through analysing future scenarios for different years and locations by planners and researchers. This integration, allowing for predictive modeling, classification and visualization within the same framework adds to usability, thus making this system powerful for practical work. In conclusion, the present approach provides a robust solution for predicting urban heat and mapping it based on data. The combined regression and Random Forest model in this framework captures both linear and nonlinear relationships, resulting in better prediction performance and practical insights relevant to sustainable urban planning in relation to heat prevention strategies.

V. Model Training and Evaluation

A hybrid prediction model using regression techniques and Random Forest algorithm is trained and tested to deliver an accurate estimate of land surface temperature (LST) as a function of several environmental and spatial drivers. In this study, NDVI (Normalized Difference Vegetation Index), NDBI (Normalized Difference Built-up Index), LULC (Land use/Land cover) and the previous records of LST are on the database employed for this work. Dataset is divided into training and testing subsets (usually 70:30 or 80:20) to validate the learning. We split our data into training and testing sets to train our predictive models and use the test set for performance assessment and generalization capabilities.Regression techniques are first used to seed the relationships between input features and temperature, primarily for linear dependency analysis to mimicking the underlying equations. Even so, given the intricacies and non-linear relationships of urban heat, Random Forest is still used as a primary predictive model. Random forest is an ensemble learning algorithm that builds several decision trees based on random subsets of the features and data samples and aggregates their output to generate a more accurate and stable prediction. This method reduces overfitting and enhances robustness, resulting in a good application for various types of environmental data. Additionally, hyperparameter tuning occurs via modifying variables including the number of trees, maximum depth of a tree, and feature selection procedure in order to achieve maximum model performance. There is also an analysis of feature importance in relation to the Random Forest technique for determining which variables affect urban heat changes.Using the trained model, predictions for land surface temperature will be made based on user-defined inputs, such as location and year.

The predicted land surface temperatures will then be compared with observed values to assess the accuracy of the predicted value. Evaluations of model performance are completed on the predictive accuracy of the model using several evaluation metrics. These metrics are: mean absolute error (MAE), which is the average error between actual and predicted land surface temperatures; root mean square error (RMSE), which is the square root of the average squared error between actual land surface temperature and predicted land surface temperature; and coefficient of determination (R^2), which indicates how well the model predicts variance in land surface temperature values. Through these metrics, we will be able to create a complete assessment of accuracy, reliability, and efficiency of the model.The predicted temperature values will also be grouped into one of three heat intensities: high, medium and low. The heat intensity will be presented with a specific colour scheme to represent the locations; red for high temperature heat spots; orange for moderate temperature areas; and green for low temperature areas. The classification of the temperature values provides greater ease of interpretation, as well as the ability to effectively validate how heat has been distributed spatially.There is a found consistency with predicted outcomes and what was actually observed so confirms that the proposed modelling method is robust. The provision of uniform training and careful parameter tuning along with multiple different methods of evaluation that are unambiguous support to produce correct urban heat forecasts, thereby providing for an operationally viable method of developing heat forecasts for cities.

VI. Experimental Results

The performance of the proposed urban heat prediction framework is evaluated using a combination of spatial visualization, temporal forecasting, and inter-city comparison. The experimental results demonstrate the effectiveness of integrating regression techniques with the Random Forest algorithm for predicting land surface temperature (LST) and analyzing urban heat patterns.

Fig. 1 presents the prediction summary for the selected city, including historical and forecasted land surface temperature values. The results indicate a gradual increase in temperature from 2010 to future projected years such as 2030 and 2038. This trend highlights the continuous impact of urbanization and environmental changes on urban heat intensity. The Random Forest model achieved high prediction performance, as indicated by strong R^2 values and low error metrics, confirming its reliability in capturing both linear and nonlinear relationships.

Fig. 2 illustrates the spatial distribution of land surface temperature using a GIS-based heat map. The visualization clearly distinguishes high-temperature regions (red), moderate regions (orange), and low-temperature areas (green). It can be observed that densely built-up urban regions exhibit significantly higher temperatures compared to areas with vegetation

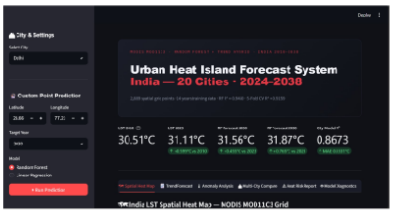

Fig. 1. Prediction summary showing historical and forecasted land surface temperature values.

or lower urban density. This spatial representation effectively identifies urban heat hotspots and supports location-based analysis.

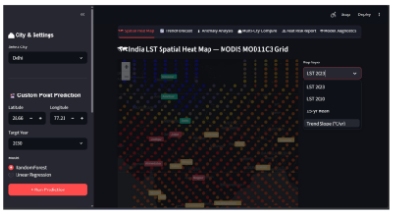

Fig. 2. Spatial heat map showing land surface temperature distribution across the study area.

Fig. 3 shows the temporal trend and forecast of land surface temperature over time. The graph demonstrates a consistent upward trend, indicating a steady increase in urban temperature. The model successfully captures both historical variations and future projections, validating its capability for long-term forecasting. This trend analysis provides important insights into the progression of urban heat and its potential future impact.

Fig. 3. Temporal trend and forecast of land surface temperature.

Fig. 4 presents a comparative analysis of predicted land surface temperature across multiple cities. The results reveal noticeable variations among different urban areas, with some cities exhibiting significantly higher thermal intensity. This comparison helps in identifying vulnerable regions and prioritizing heat mitigation strategies based on city-level analysis.

Fig. 4. Comparison of land surface temperature across multiple cities.

Overall, the experimental results confirm that the proposed framework provides accurate and reliable predictions of urban heat patterns. The integration of machine learning-based forecasting, spatial heat mapping, and comparative analysis enhances the interpretability of results and supports effective decision-making for sustainable urban planning and heat mitigation strategies.

A comparative analysis of the models indicates that the Random Forest algorithm outperforms the regression model in terms of prediction accuracy and reliability.

TABLE II
COMPARISON OF REGRESSION AND RANDOM FOREST MODELS

Model	MAE	RMSE	R^2
Regression	1.25	1.80	0.82
Random Forest	0.85	1.20	0.91

A. Discussion

Results from an urban heat prediction system show that this model can capture spatial and temporal variations of land surface temperature effectively, which is done by combining regression modelling with Random Forest algorithms enabling modelling to complex how environmental parameters like vegetation/built up area have many relationships with temperature.Forecast results show that temperature has gradually increased over time indicating that rapid urbanization and environmental change are responsible for the increase in heat. The system's capability to define input parameters (location/year) means it can be useful for making predictions of urban heat for forecast periods based on different geographic regions.

The heat map shows a clear display of the location of people/temperature distribution in all geographic areas listed. Higher temperature areas of considerable density are mainly found in dense urban environments, whereas areas characterized by significant volumes of vegetation typically have lower temperature overall. These results confirm that vegetation and land use types have a major impact on urban heat formation.In addition, a trend analysis supports these results by identifying

an increase in temperature over time. As such, there is a growing need for heat mitigation strategies and effective urban planning. Additionally, different cities have vastly different temperature levels depending on where they are located and whether they have properly built up their infrastructure and plants.

A key limitation of the model is that the accuracy of predictions is determined by the input data. Therefore, if input data have insufficient technical quality, predictions will also be inaccurate. For example, predicting temperatures during a natural disaster or when the area experiences a severe weather event may not be accurate using this model. Adding additional characteristics to the model, including humidity, population density, and traffic data, could improve the accuracy of the predictions.The system proposed in this report is easy to use, practical, and scalable. It will provide urban planners and policymakers with valuable information and ways to help them better understand and control urban heat, and therefore is a valuable resource for sustainable urban development.

VII. Conclusion

A successful integrated method for urban heat prediction through regression and random forests is presented. The method combines ML, geospatial, and visualization techniques for predicting land surface temperature (LST) and locating urban heat patterns.

The results indicate that the model is capable of accurately capturing spatial and temporal temperatures. The use of multiple environmental factors including vegetation, built-up areas, and land use provides greater accuracy of prediction. By categorizing LST into high, medium, and low using colour coding, this enhances interpretability and makes the model more user-Friendly.Overall, urban temperature data analysis suggests an upward trend over time, indicative of the increasing influence of urbanization. The spatial heat maps, along with the city-to-city analyses, provide information regarding heat distribution in cities and suitable locations for locating urban areas at high risk for heat impacts. This shows the need for data-driven methods to describe and to manage heat in urban environments.

Overall, the proposed approach is effective, scalable, and appropriate for use in the real world. Urban planners and policymakers can utilize these results to develop heat mitigation strategies while fostering sustainable urban development. Future research could extend this work by including more variables (e.g., humidity, population density, traffic) to improve predictive capabilities and expand overall applicability.

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